Harshit Kumar

Bachelor of Technology in Computer Science Engineering | IIIT Kota, Rajasthan
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SUMMARY

Third-year CSE student at IIIT Kota (2024–2028) specializing in scalable GenAI systems, multi-agent AI orchestration, and high-throughput backend APIs. Proficient in C++, Python, FastAPI, LangChain, and LangGraph. Delivered production-ready systems processing 10,000+ daily queries at sub-200ms latency, committed to engineering highly optimized and resilient digital infrastructure.

EDUCATION

Indian Institute of Information Technology, Kota *Aug 2024 – May 2028 (Expected)*
Bachelor of Technology in Computer Science Engineering

SKILLS

Languages: C++, C, Python, Java, JavaScript, TypeScript
AI / ML: LangChain, LangGraph, Groq LLM, Gemini API, Hugging Face, Cerebras, Agentic AI, Multi-Agent Systems, Prompt Engineering, NLP, Geospatial Analytics
Backend & Architecture: FastAPI, Node.js, RESTful APIs, System Design, Microservices **Libraries:** NumPy, Pandas, Matplotlib
Databases: SQL, MongoDB, SQLite, ChromaDB **Tools:** Docker, Git, GitHub, Chrome Extension (Manifest V3)

PROJECTS

Basera & Homie – Direct Relocation Engine GitHub | Live | *May 2026 – Jun 2026*
AI-powered housing platform with virtual LLM landlords and real-time geospatial search
- Engineered a real-time LLM negotiation simulator using Groq/Llama-3 streaming APIs, achieving under 350ms average response latency and minimizing time-to-first-token (TTFT).
- Built custom point-in-polygon ray-casting algorithms for geospatial mapping across 1,000+ coordinates achieving under 8ms execution time.
- Eliminated cold-start DB connection overhead by 92% (2,000ms to under 150ms) via a custom DNS-lookup patch, and optimized MongoDB compound indices.

TruthShield – Deepfake Detection Platform GitHub | Live | *May 2026 – Jun 2026*
High-throughput security API for real-time visual forensics and text scam identification
- Architected a scalable detection platform integrating Hugging Face CV models and Groq LLMs, achieving approximately 92% overall accuracy (peaking at 94% for text diagnostics).
- Developed FastAPI and Node.js REST APIs minimizing text inference latency to under 120ms and complex visual forensics to under 1.8 seconds on concurrent data streams.
- Deployed robust backend infrastructure using SQL and the Gemini API, designed to scale and process 10,000+ verification queries daily with strict IP rate-limiting.

Lexis AI – Multi-Agent Content Pipeline GitHub | Live | *Feb 2026 – Mar 2026*
Modular content orchestration engine with RAG fact-checking and parallel C++ translation engine
- Architected an 8-stage stateful LangGraph pipeline integrating Groq LLMs, local ChromaDB vector store (`all-MiniLM-L6-v2`), and web search for 5-layer compliance auditing and self-correcting refinement.
- Engineered a high-performance C++ localization engine for post-approval parallel translations and integrated Stability AI APIs for automated social asset generation.
- Deployed a FastAPI backend serving a modular SPA featuring animated 3D geodesic canvas visuals; containerized via Docker with complete Pytest automation coverage.

LEADERSHIP

Social Media Lead – Esports Club Clutch, IIIT Kota *Aug 2024 – Present*
- Grew club's social media following by 30%+ through targeted content planning and consistent community engagement.
- Created promotional content and managed online marketing campaigns for inter-college esports tournaments.

Senior Executive Event Manager – TechKnow Council, IIIT Kota *Jan 2025 – May 2026*
- Planned and executed 5+ technical events, hackathons, and workshops managing a total footfall of over 6,000 attendees.
- Coordinated 10+ vendors and cross-functional teams end-to-end across budgeting, sponsorships, registrations, and on-ground operations.
- Managed corporate sponsorship operations and formulated targeted partnership proposals for Tech Summit 2026, featuring the HackTheChain 4.0 hackathon.

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** Data Analytics Job Simulation – Deloitte (Jun 2026); AI Foundations – ISTE & IBM.
- Solved 300+ DSA problems using C++ across coding platforms; achieved 2-Star rating on CodeChef under strict time and memory constraints.
- Qualified Phase 2 of the ET Gen AI Hackathon by Economic Times, ranking in the top 6,000 of 54,000+ participating teams.